

Standard Model of Elementary Particles + Gravity

three generations of matter
(fermions)

interactions / force carriers
(bosons)

I

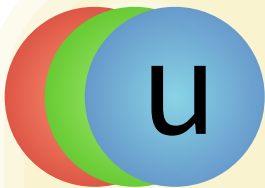
II

III

mass
charge
spin

$\approx 2.4 \text{ MeV}/c^2$

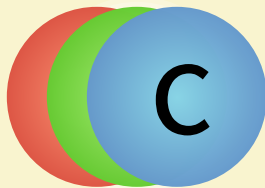
$2/3$
 $1/2$



up

$\approx 1.275 \text{ GeV}/c^2$

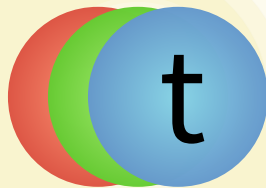
$2/3$
 $1/2$



charm

$\approx 172.44 \text{ GeV}/c^2$

$2/3$
 $1/2$



top

$\approx 4.8 \text{ MeV}/c^2$

$-1/3$
 $1/2$



down

$\approx 95 \text{ MeV}/c^2$

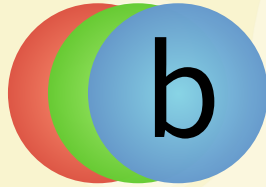
$-1/3$
 $1/2$



strange

$\approx 4.18 \text{ GeV}/c^2$

$-1/3$
 $1/2$



bottom

$\approx 0.511 \text{ MeV}/c^2$

-1
 $1/2$



electron

$\approx 105.67 \text{ MeV}/c^2$

-1
 $1/2$



muon

$\approx 1.7768 \text{ GeV}/c^2$

-1
 $1/2$



tau

$< 2.2 \text{ eV}/c^2$

0
 $1/2$



electron
neutrino

$< 1.7 \text{ MeV}/c^2$

0
 $1/2$



muon
neutrino

$< 15.5 \text{ MeV}/c^2$

0
 $1/2$



tau
neutrino

0
 0
 1



gluon

α_s

0
 0
 1



photon

α

$\approx 91.19 \text{ GeV}/c^2$

0
 1



Z boson

α_w

$\approx 80.39 \text{ GeV}/c^2$

± 1
 1



W boson

α_w

Coupling Constants
Strong $\alpha_s \approx 1$
EM $\alpha = 1/137$
Weak $\alpha_w \approx 10^{-6}$
Gravity $\alpha_g \approx 10^{-39}$

0
 0
 2



graviton

α_g

QUARKS

LEPTONS

GAUGE BOSONS
VECTOR BOSONS

SCALAR BOSONS

HYPOTHETICAL
TENSOR BOSONS

$\approx 125.09 \text{ GeV}/c^2$

0
 0
 0



higgs